

Job Title: Motor Controller Application Engineer
Department: Powertrain / Technology

Job Summary:-

We are seeking a highly motivated and skilled Motor Controller Application Engineer to join our Electric Vehicle development team. The ideal candidate will be responsible for integration, calibration, and validation of motor controllers across various EV platforms. This role requires deep technical understanding of motor control algorithms, CAN communication, and real-world vehicle application tuning.

Key Responsibilities:

- Lead the integration and validation of motor controllers with vehicle systems.
- Work closely with firmware, electrical, and mechanical teams to ensure seamless system compatibility.
- Define and calibrate motor control parameters for optimal performance, efficiency, and thermal management.
- Interface with controller suppliers for software customization, bug resolution, and feature upgrades.
- Conduct vehicle-level testing and tuning for different drive modes (Eco, Sport, Regen, etc.).
- Analyze CAN logs and develop automated scripts for performance evaluation and debugging.
- Support in development of power maps, torque curves, and regen profiles as per vehicle requirement.
- Validate motor control functions under various road and load conditions.
- Assist in defining functional safety and diagnostic requirements in compliance with ISO 26262.
- Collaborate with cross-functional teams to support homologation and regulatory testing.

Required Skills & Qualifications:-

- Bachelor's or Master's degree in Electrical / Electronics / Mechatronics Engineering.
- relevant college experience in motor control systems (BLDC / PMSM / SRM).
- Strong understanding of motor control theory, field-oriented control (FOC), sensor calibration, and tuning.
- Experience with CAN tools (e.g., Vector CANalyzer, CANoe, PCAN) and CAN protocol analysis.
- Proficiency in MATLAB/Simulink, Python, or similar tools for data processing and parameter tuning.
- Familiarity with vehicle-level testing, thermal derating strategies, and protection mechanisms.
- Understanding of vehicle architecture, HV systems, and functional safety standards.

Preferred Experience:-

- fresher/ Experience working with EV OEMs or Tier-1 motor/controller suppliers.
- Exposure to motor inverter hardware and diagnostic protocols (UDS)
- Excellent problem-solving skills and ability to work in a fast-paced development environment.

- Strong documentation and technical communication skills.